Lirui (Leroy) Wang

(206) 483-8515 • liruiw@uw.edu • 5000 25th Ave NE, Seattle, WA 98105
http://www.linkedin.com/in/lirui-wang • https://github.com/liruiw

Education:

University of Washington, Seattle, WA (GPA:**3.96**/**Annual Dean's list**)

- Bachelor of Science, Electrical Engineering (Honor), College of Engineering — Expected Jun 2020
- Bachelor of Science, Computer Science (Honor), College of Arts and Sciences — Expected Jun 2020
- Bachelor of Science, Mathematics (Minor), College of Arts and Sciences — Expected Jun 2020

Work Experience:

Intern, DJI Technology, RoboMasters Department, Shenzhen, China — Jul 2017 – Aug 2017

- Worked on a robot and a drone (with Nvidia TX1 onboard) that would compete in a fully automatic robotic battle
- Programmed take-off platform, barrier tape and tag detection for the drone to retrieve info such as relative location
- Applied probabilistic algorithm to navigate and locate, and built a top-level state machine to strategize our robot

Projects:

Grasping Project – **UW Robotics and States Estimation Lab** — Dec 2018 – Present

- Apply traditional package GraspIt! to a table-top object pick up tasks
- Design new learning-based approach that do grasp pose synthesis based on geometry only

DeepIM Tracking Project – **UW Robotics and States Estimation Lab** — Aug 2018 – Dec 2018

- Do extensive experiments of DeepIM on YCB datasets and achieves good results trained on synthetic data only
- Improve network training stability, pose distribution and occlusion generation
- Apply the tasks in real-time tracking for robot hands and AR demos

Robotic Alignment Project – **UW Robotics and States Estimation Lab** — June 2018 – Aug 2018

- Attempt to refine the SE3 poses of robot by training the network on a real-time synthetic manner
- Use computer graphics tools and 3D geometrics to calculate and perturb poses for training
- Work on kinematics to build generalized system that applies to robot systems

Depth and Flow Research – **Computer Vision Course Project** — March 2018 – June 2018

- Reproduced some deep learning methods results for depths and flow predictions and evaluated their performances.
- Made a poster on structures and ideas such as SfMlearner and MonoDepth and trained and tested them on a robotic setting dataset.

SoundScape Project – **UW Adaptive Computing Machines and Emulator Lab** — March 2018 – June 2018

- Designed a sound processing pipeline from scratch for modelling the acoustics of a train whistle.

Optical Manipulation Project – **UW Robotics and States Estimation Lab** — Aug 2017 – March 2018

- Worked as a research assistant to implement a manipulation demo for PoseCNN, which enables the robot Baxter to classify and manipulate objects.
- Studied camera models, robot kinematics, Baxter SDK and ROS packages for tracking.

Leadership and Involvement:

Advanced Robotics at the University of Washington – Computer Vision Lead & Drone Lead — Sep 2016 – Feb 2018

- Managed a team of 10 engineering students to design computer vision for our robots in a robotic competition RoboMasters hosted by DJI, one of the world's current leader in drone manufacturing
- Used Python, ROS, C++ to perform system integration of controls, navigation and serial communication
- Implement the detection, shooting, refilling, landing and communication using DJI SDK on DJI M100 drone

Skills:

Languages: Java, C++, C, Python
Platform & Frameworks: ROS(Robot OS), Ubuntu, ARM microcontrollers, Arduino, Nvidia TX Series, STM32
Tools: Matlab, Git, CMake, OpenCV, DJI_Guidance, Bash, Latex, Tensorflow, Pytorch, MXNet, OpenGL, CUDA
Courses: Computer Vision, Deep Learning, Artificial Intelligence, Graphics, Robotics, Algorithms, Numerical Analysis, Linear Analysis, Convex Analysis and Optimization, Embedded Systems, Control Theory, Digital Signal Processing